6 Fig. 6.1 shows a road next to the sea.



Fig. 6.1

(a) On a sunny day, the Sun warms the road.

Describe how energy from the Sun reaches the Earth and warms the road.

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..... [3]

(b) The temperature of the road is greater than the temperature of the sea.

The surface of the road is black.

Suggest **one** reason why the temperature of the road is greater than that of the sea.

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..... [1]

(c) The air above the road is heated by the warm road.

(i) Describe how this affects the molecules of the air.

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..... [2]

(ii) A cyclist travelling along the road notices that a cool breeze is blowing from the sea to the land.

Explain how convection produces this breeze. You may include a diagram if it helps your answer.

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..... [3]

[Total: 9]